

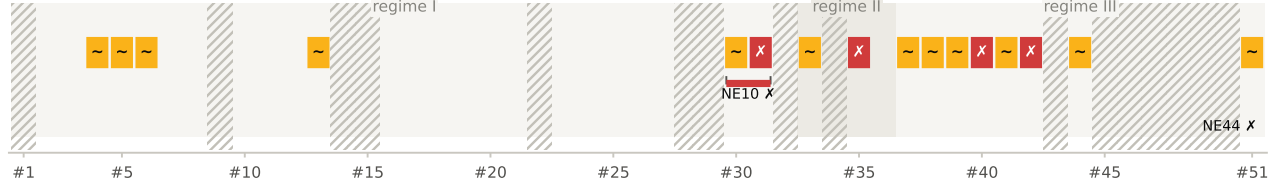
# H30: An operator-susceptibility gauge $\chi_{\text{op}}$ : how steerable is the swarm today?

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 2/5 Usefulness 2.5/5

**Bottom line.** Operator messages have a period-level susceptibility (about one active minute per nudge; a small content pull), but daily steerability is not a trackable state.

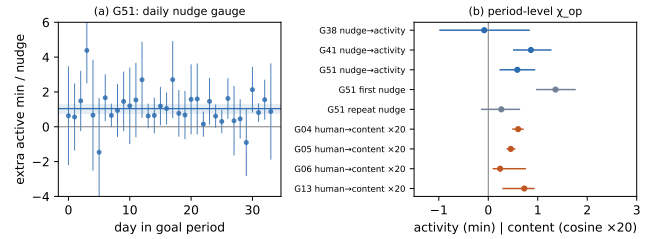
level: hypothesis · pipeline checks 86% · updated 2026-10-04



**Goal periods tested (6 failed, 12 mixed).** ✓ supported ✗ failed ~ mixed d descriptive ... pending.  
Heavy outline: confirmatory test on the locked holdout. Hatched columns: held-out goal periods. Brackets: tests spanning a natural experiment.

## WHAT HAPPENED

We measure how far one operator message moves the agents who see it, and whether that number is trackable by day. We test 16 non-holdout goal periods, after a synthetic validation that fixed three estimator biases. **Round 1b** uses the corrected activity table, because round 1 lost half of all events. It adds the leading-@ nudge target, receiving-call timing and past-only controls with day effects. *Activity*: a nudge adds about one active minute for its target (#51 0.98 [0.63, 1.37], #38 0.95 [0.38, 1.54]; round 1: 0.59, -0.08). First nudges give 1.21, repeats 0.44, room-mates  $\approx 0$ . The round-1 lull warning reverses (lull 1.74 vs 0.72). *Content*: human messages turn the next statements toward their new content, +0.024 (room-mates) and +0.041 (named), in both embedding models. *Daily gauge*: #51 varies by day ( $p = 0.026$ ,  $R_1 = 0.48$ ) without persistence. NE44: nudges buy more under the 5-min pause default (reversed). The holdout is not run.



**Observables.** (a) #51 daily nudge gauge: extra active minutes per nudge for the target (day FE, 95% CIs), with the period mean and band. Day-to-day scatter matches the daily CIs: no trend, no persistence. (b) Period-level  $\chi_{\text{op}}$ . Nudge  $\rightarrow$  activity in the three largest nudge periods (pre-registered model); first vs repeat nudges in #51 (day FE); human  $\rightarrow$  content in human-dense periods (cosine  $\times 20$ ).

## SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Do not watch a daily “steerability” number: at village message rates it is a period constant plus noise, and it does not age. One nudge to an idle agent buys about one extra active minute from its receiving call. A first nudge buys 1.2 min and a repeat 0.4 (CI includes 0). Nudging into a swarm-wide lull works even better (1.74 vs 0.72 min), and short scaffold pauses help (NE44). To change *what* an agent writes, name it and add new content (+0.041 vs +0.024 for room-mates). For analysts: do not condition on future nudges, as H04’s isolation rule did.

## PHYSICAL MODEL

Linear response to impulsive fields (01, 11). Each operator message  $k$  is a field pulse on its recipients  $i$ . Spins: activity  $n_i(t) \in \{0, 1\}$ ; content  $s_i$ , a whitened statement embedding.  $u_k^\perp$ : the message direction projected off the recipient’s last 8 statements.  $\chi_{\text{act}}$ : the 30-min integrated activity response (local projection; strata  $\alpha_s$ , day effects  $\delta_d$ , kick counts  $K$ ).  $\chi_{\text{con}}$ : post-message alignment minus same-kind placebos  $p$ .  $\chi_{\text{op}}(d)$ : the day mean of per-kick responses  $r_k$ .

$$Y_i(m) \equiv \sum_{\tau=1}^{30} n_i(m + \tau) = \alpha_{s(i,m)} + \delta_d + \sum_c \chi_{\text{act}}^c K_i^c(m) + \beta \cdot K_i^{\text{adj}}(m) + \varepsilon$$

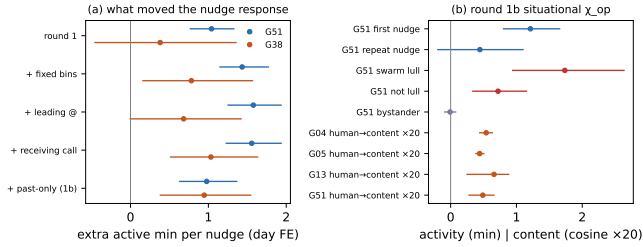
$$\chi_{\text{con},k} = \bar{s}_i^{\text{post}} \cdot u_k^\perp - \langle \bar{s}_i^{\text{post}} \cdot u_p^\perp \rangle_p, \quad u^\perp = \frac{(1 - P_{\text{pre}})u}{\|(1 - P_{\text{pre}})u\|}$$

$$\chi_{\text{op}}(d) = \frac{\sum_{k \in d} w_k r_k}{\sum_{k \in d} w_k}, \quad R_1 = 1 - \frac{\text{Var}_{\text{perm}}[\chi_{\text{op}}(d)]}{\text{Var}[\chi_{\text{op}}(d)]}$$

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## RESULTS VS PREDICTIONS

| Prediction           | Round 1 → round 1b                                                                  | Verdict          |
|----------------------|-------------------------------------------------------------------------------------|------------------|
| P1 nudge→target      | #51 0.59 → 0.98 [0.63, 1.37]; #38 -0.08 → 0.95 [0.38, 1.54]; small periods 5/10 > 0 | ✓ (count rule ×) |
| P2 bystanders ≈ 0    | #51 0.13 → -0.01; #38 -0.54 → -0.09                                                 | ✓                |
| P3 human→activity    | ≤ 0 / n.s. in regime I (unchanged)                                                  | ×                |
| P4 human→content     | 0.026 → 0.024 (gte 0.024); named 0.046 → 0.041                                      | ✓                |
| P5 nudge→content     | #51 0.003 → 0.002 [-0.001, 0.005]                                                   | ✓                |
| P6 no daily state    | #51 $p$ 0.08 → 0.026, $R_1$ 0.28 → 0.48, lag-1 0.08                                 | × as written     |
| P7 no aging          | #51 +0.02 [-0.03, 0.07]/day                                                         | ✓                |
| P8 context saw-tooth | pull falls 0.66 vs late 1.49                                                        | ×                |
| P10 family           | no Holm-significant lab                                                             | ✓                |
| P11 nulls            | #51 and #38 now pass                                                                | ✓                |
| Native #5 dose       | #51 +0.02 → 0.013 over 1–7 msg; 8+ noisy                                            | × (rule)         |
| Native NE44          | 12-h 0.57 vs 5-min 0.98 ( $\Delta$ -0.41)                                           | × reversed       |
| Native NE10          | first nudges ever -0.42 (22 kicks)                                                  | × low power      |



**Second figure.** Round 1b. (a) What moved the nudge response in #51 (blue) and #38 (orange). Restoring the dropped events carries most of the change; past-only controls bring the level back to  $\approx 1$  min. (b) Situational  $\chi_{op}$  in #51 (first vs repeat nudges, swarm lull vs not, bystanders) and human content pull ( $\times 20$ ).

## GOAL PERIODS IN DETAIL

G04  $\sim$  1b: mixed (r1 mixed; fixed bins, receiving call)  
G05  $\sim$  1b: mixed (r1 mixed) · native dilution: failed  
G06  $\sim$  1b: mixed (r1 mixed; fixed bins, receiving call)  
G13  $\sim$  1b: mixed (r1 mixed; fixed bins, receiving call)  
G30  $\sim$  1b: mixed (r1 mixed; fixed bins, receiving call)  
G31  $\times$  1b: failed (r1 mixed; fixed bins, receiving call)  
G33  $\sim$  1b: mixed (r1 mixed; fixed bins, receiving call)  
G35  $\times$  1b: failed (r1 failed; fixed bins, receiving call)  
G37  $\sim$  1b: mixed (r1 mixed; fixed bins, receiving call)  
G38  $\sim$  1b: mixed (r1 mixed; fixed bins, receiving call)  
+8 more periods: see the card.

## FAITHFULNESS SCORECARD

|                        |   |                   |   |
|------------------------|---|-------------------|---|
| A mapping              | 1 | F identifiability | 1 |
| B assumptions          | 1 | G ground truth    | 1 |
| C adequacy             | 1 | H comparative     | 1 |
| D unfitted predictions | 1 | I transfer        | 1 |
| E interventional       | 0 |                   |   |

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

## CAVEATS

Only #51 and #38 resolve the nudge response; 3–5-day periods give unstable CIs. The level depends on the future-kick adjustment: 0.98 past-only vs 1.56 with the A1 regressors of round 1 (day FE). SE-based window reliabilities exceed 0.5 in #51 but also in #38, where the calibrated permutation finds no day signal, so we do not count them. Human effects on activity in dense regime-I chat are not identified. Content measures message-specific steering only, and human replies may share topics. The round-1b design (receiving-call timing, past-only, day FE primary) follows the re-evaluation brief, not the pre-registration.

## ROUND 2

- **Essence.** a trustworthy answer to "will my next message change what this swarm does?". Round 1 says it is per situation, not a daily state.
- **H30-R1** Situational  $\chi$  instead of daily  $\chi$ .
- **H30-R2** The 2026-08-20 switch-off as a natural experiment.
- **H30-R3** Embedding swap and reply control for  $\chi_{con}$ .
- **H30-R4** Per-message content strength.
- **H30-R5** Run `confirm.py` after the card is committed.

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